Department of Physics, Princeton University

Graduate Preliminary Examination
Part I

Thursday, January 21, 2021
9:00 am - 12:00 noon

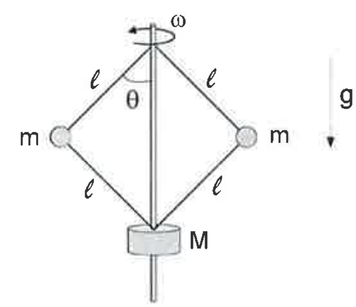
Answer TWO out of the THREE questions in Section A (Mechanics) and TWO out of the THREE questions in Section B (Electricity and Magnetism).

Work each problem in a separate examination booklet.

Be sure to label each booklet with your name, the section name, and the problem number.

Section A. Mechanics

1. **Flyball governor**



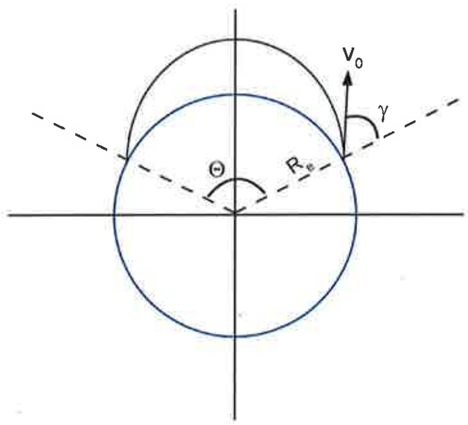
A flyball governor is a kind of device used to regulate motor speed. A simplified version of this device is shown in the figure. It consists of a rotating shaft connected to two equal masses m by massless rigid rods of length ℓ. The rods are also attached at the bottom to a mass M that can freely slide along the vertical shaft. You may treat all masses as pointlike, and there is no friction. The uniform gravitational field is along the vertical direction as shown in figure, and the angular velocity ω is constant.

(a) Suppose you observe the governor to be at an equilibrium angle θ_0. Find the expression for the equilibrium angle as a function of the angular velocity ω and the other parameters in the problem.

(b) What is the minimum angular velocity ω such that the equilibrium angle θ_0 is non-zero? What is the behavior of the equilibrium angle in the limit $\omega \to \infty$?

(c) Find the frequency of small oscillations of the angle θ about the equilibrium position θ_0 (assuming ω is larger than the minimum required angular velocity determined in the previous question) .

2. **Ballistic missile**



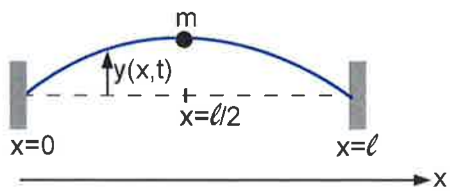
A ballistic missile accelerates quickly from rest and then coasts until it hits the surface of the earth. A good approximation is to treat it like a projectile with some initial speed v_0 whose subsequent motion is determined by Earth's gravitational field alone. You may neglect air resistance and the rotation of the Earth.

Suppose we launch a ballistic missile at an angle γ from vertical and at an initial speed v_0. A convenient measure of how far the missile goes is the arc-distance Θ, see the figure. It is convenient to define $\eta = v_0^2/(gR_e)$, where R_e is the radius of the Earth, and g the gravitational acceleration at Earth's surface. (You may not assume that the gravitational field is uniform in this problem!)

(a) Find the launch angle γ that results in the maximum arc-distance Θ. Express your result for the launch angle γ and the corresponding maximum Θ in terms of the variable η defined above.

(b) It is always a good idea to check your result in simple limits. Taking the limit $R_e \to \infty$, verify that your answer to the previous question reduces to the familiar result for projectile motion on a flat Earth with uniform gravitational field.

3. **Point mass on a string**



Consider a string of length ℓ with uniform mass density σ and tension τ (both assumed constants), and endpoints at $x = 0$ and $x = \ell$ held fixed. A point mass m is attached at the center of the string, at $x = \ell/2$. The mass can only move in the vertical direction (it cannot slide along the string). Gravity can be neglected. The purpose of this problem is to find the normal mode frequencies for the small vertical oscillations $y(x,t)$ of the string in the presence of the mass. You may assume that $y(x,t)$ obeys the standard wave equation

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2}, \qquad v^2 = \frac{\tau}{\sigma}.$$

(a) The wave equation should be supplemented with the boundary conditions at $x = 0$, $x = \ell$ (which should be obvious), and with appropriate "matching conditions" on $y(x,t)$ and its derivative at $x = \ell/2$. Determine the form of these matching conditions.

(b) We may search for solutions of the wave equation in the normal mode form

$$y(x,t) = \rho(x)\sin(\omega t + \phi)$$

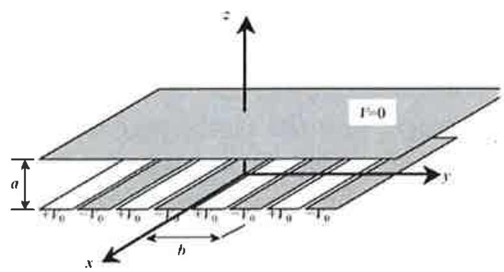
Use the boundary conditions and matching conditions derived above to find the possible oscillation frequencies ω. It is convenient to consider separately the modes for which the string has a node at $x = \ell/2$ ("odd modes") and those for which it does not have a node at $x = \ell/2$ ("even modes"). For the odd modes, determine the form of the frequencies explicitly. For the even modes, find the transcendental equation that determines the allowed frequencies.

(c) Find the form of the even mode frequencies in the limit $m/(\sigma\ell) \to 0$.

(d) In the opposite limit $m/(\sigma\ell) \to \infty$, find an approximate expression for the lowest oscillation frequency, to leading non-trival order in the small quantity $\sigma\ell/m$.

Section B. Electricity and Magnetism

1. **An electrostatics problem**



A device is made of two very large conducting parallel planes separated by a distance a in the z direction. The distance a is much smaller than the size of the planes, which you may take to be infinite in this problem. The plane at $z = a$ is grounded, and you can set its electrostatic potential to be $V = 0$. The plane at $z = 0$ is made of parallel strips of width $b/2$, see the figure. The voltage on the strips is alternating between $+V_0$ and $-V_0$. You may ignore the gaps between the strips.

Find the electrostatic potential $V(y, z)$ everywhere between the plates. Your final result can be left in the form of an infinite series. [Hint: Use separation of variables, and also note that the voltage of the lower plane can be regarded as a periodic function of y with period b.]

2. **A sudden burst of current**

A sudden burst of current flows in a straight infinite wire. For the purpose of this problem, we may describe this in a simplified way as an instantaneous current burst at $t = 0$ given by

$$I(t) = q_0\delta(t)$$

where $\delta(t)$ is the Dirac delta-function, and q_0 a constant with units of electric charge. You may assume that the wire is electrically neutral.

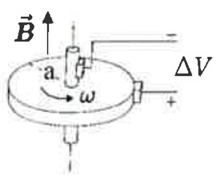
(a) Working in the Lorentz gauge, find the electromagnetic potentials as a function of time and position. It is convenient to use cylindrical coordinates z, r, ϕ, with z along the wire, r being the transverse distance to the wire, and ϕ the angle around the wire.

(b) Find the electric field $\vec{E}(\vec{r}, t)$ and the magnetic field $\vec{B}(\vec{r}, t)$.

(c) Consider a cylindrical surface of radius R and height h centered at the wire, surrounding it. What is the total electromagnetic power flowing through this surface as a function of time?

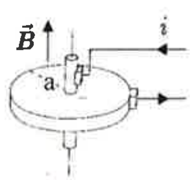
3. **Homopolar generator**

A "homopolar generator" consists of a conducting disk of radius a that can rotate about an axis through its center, in the presence of a uniform magnetic field of magnitude B in the vertical direction, as shown in the figure. A wire is connected to the shaft through a frictionless contact, and another wire is connected at the outer edge of the wheel. Throughout the problem, you may assume that the disk is perfectly conducting, and ignore the thickness of the shaft.

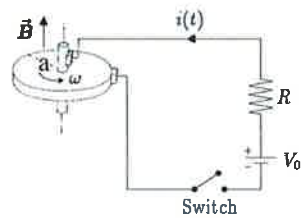


(a) If the wheel is forced to rotate with angular velocity ω (counterclockwise as viewed from the top, as in the figure above), a potential difference ΔV is generated between the shaft and the outer edge of the disk. Find ΔV as a function of B, a, ω.

(b) If a current i flows through the wheel, radially from the center to the edge as shown in the figure below, a torque τ along the axis is exerted on the wheel (with positive τ meaning a counterclockwise angular acceleration). Calculate τ (both its magnitude and sign) as a function of B, a and i.



(c) Suppose that the wheel is connected to a circuit including a DC voltage generator V_0, a resistance R and a switch. Neglect all other sources of resistance, as well as the self-inductance of the system. Denote by I_0 the moment of inertia of the wheel about its axis. The switch is open with the wheel at rest for $t < 0$. At $t = 0$, the switch is closed. Find the current $i(t)$ and the angular velocity $\omega(t)$ of the wheel.



Department of Physics, Princeton University

Graduate Preliminary Examination
Part II

Friday, January 22, 2021
9:00 am - 12:00 noon

Answer TWO out of the THREE questions in Section A (Quantum Mechanics) and TWO out of the THREE questions in Section B (Thermodynamics and Statistical Mechanics).

Work each problem in a separate booklet. Be sure to label each booklet with your name, the section name, and the problem number.

Section A. Quantum Mechanics

1. **Harmonic oscillator in electric field**

   A particle of mass m and electric charge q moves in one dimension under the influence of a harmonic potential and a uniform electric field. The Hamiltonian is

   $$H = \frac{p^2}{2m} + \frac{m\omega^2}{2}x^2 - q\mathcal{E}x$$

   where $\mathcal{E}$ is a constant.

   (a) Show that the Hamiltonian can be written as $H = e^{-iA}H_0e^{iA} + B$, where A and B are Hermitean operators, and $H_0 = \frac{p^2}{2m} + \frac{m\omega^2}{2}x^2$ is the Hamiltonian unperturbed by the electric field. Find A and B explicitly.

   (b) Use the result of the previous question to find the energy spectrum of H, and to express the energy eigenstates of H in terms of those of H_0. You may denote as $|n;\mathcal{E}\rangle$ the eigenstates of H, and just as $|n\rangle$ those of H_0 ($|n\rangle \equiv |n;\mathcal{E}=0\rangle$).

   (c) Suppose that the particle is in the ground state of H. What is the expectation value of the position operator?

   (d) Suppose now that the particle is found at time $t = 0$ to be in the ground state of H_0 (*not* H!). What is the probability that it is in the ground state of H at time t?

   (e) Still assuming that the particle is in the ground state of H_0 at time $t = 0$, what is the probability of finding the particle again in the same state at time t? For what times t is this probability maximum?

   (f) Again assuming that the oscillator is in the ground state of H_0 at $t = 0$, find the expectation value of the position operator x as a function of time.

2. **Poschl-Teller potential**

Consider a particle of mass m moving in one-dimension under the influence of the potential

$$U(x) = \frac{\hbar^2 a^2}{2m}\left(1 - \frac{2}{\cosh^2(ax)}\right) \tag{1}$$

This is a special case of the so-called Poschl-Teller potentials, which are exactly solvable. The goal of this problem will be to show that this potential admits precisely one bound state (the ground state) with energy $E = 0$, using some ideas from supersymmetric quantum mechanics (you do not need to know anything about supersymmetry in order to solve the problem).

(a) Consider first the following general situation. Suppose you have two Hamiltonians, one of the form $H = A^\dagger A$, and another one given by $\tilde{H} = AA^\dagger$, for some operator A. Show that, assuming H has a set of normalized eigenstates $H|n\rangle = E_n|n\rangle$, then it must be that $E_n \geq 0$. Further, show that for each normalized eigenstate $|n\rangle$ of H with nonzero energy E_n, there is a corresponding normalized eigenstate $|\tilde{n}\rangle$ of $\tilde{H}$ with the same energy. Derive the explicit form of $|\tilde{n}\rangle$ in terms of $|n\rangle$. This shows that the nonzero energy eigenstates of H and $\tilde{H}$ are in one-to-one correspondence.

(b) Show that, taking $A = \frac{\hbar}{\sqrt{2m}}\frac{d}{dx} + W(x)$, both H and $\tilde{H}$ have the standard form

$$H = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + U(x)\,, \qquad \tilde{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \tilde{U}(x)$$

Determine the explicit relation between $U(x)$, $\tilde{U}(x)$ and $W(x)$.

(c) Now consider the Hamiltonian for a point mass m in the potential in eq. (1). Show that it can be written as $H = A^\dagger A$, with $A = \frac{\hbar}{\sqrt{2m}}\frac{d}{dx} + W(x)$, and $W(x) = \frac{\hbar a}{\sqrt{2m}}\tanh(ax)$. Then, use this result to show that the Hamiltonian has a zero energy bound state satisfying $A|\psi_0\rangle = 0$ (which must be the ground state since $E \geq 0$ as you proved above). Solve for the wavefunction $\psi_0(x)$ explicitly up to overall constant. You do not need to normalize $\psi_0(x)$, but show that it is normalizable.

(d) For the form of A and $W(x)$ you found in the previous question, what is then the "dual" Hamiltonian $\tilde{H} = AA^\dagger$? Show that the form of $\tilde{H}$ implies that H cannot have any bound states with nonzero energy. Thus, this shows that the $E = 0$ ground state found above is the only bound state that the Poschl-Teller potential (1) admits.

3. **Electron-positron system**

The spin-dependent part of the Hamiltonian for an electron-positron system in the presence of a constant magnetic field in the z direction is given by

$$H = \frac{A}{\hbar^2}\vec{S}_1 \cdot \vec{S}_2 + \frac{eB}{m_e c}\left(S_{1z} - S_{2z}\right)$$

where A is a constant with units of energy and B is the magnitude of the magnetic field. The subscripts '1' and '2' refer to electron and positron, respectively.

(a) Find the energy levels and eigenstates of the Hamiltonian in the case $B = 0$.

(b) Assuming the magnetic field is weak, use perturbation theory to find the corrected energy levels to second order in the small parameter $\beta \equiv \frac{eB\hbar}{Am_e c}$, and the eigenstates to first order in β.

(c) Find the exact energy levels and eigenstates. Check your result by comparing with the calculation you did above using perturbation theory.

Section B. Statistical Mechanics and Thermodynamics

1. **One-dimensional random walk**

   Consider a random walker on a one-dimensional lattice with lattice spacing d. Every time increment Δt the walker makes a step $s_i = \pm d$ with probabilities $p_+ = p_- = 1/2$. Thus, in the limit of d small and $N = t/\Delta t \gg 1$, the distance from the origin at time t is $x(t) = \sum_i s_i$ and can be seen as a continuous variable.

   (a) Show that the time-dependent probability density $P(x,t)$ is given by

   $$P(x,t) = \frac{C}{\sqrt{t}} e^{-f\frac{x^2}{t}}, \tag{2}$$

   and determine the constants C and f as functions of d and Δt.

   (b) Show that $P(x,t)$ is a solution of the diffusion equation and determine the diffusion constant D.

   (c) Determine the mean $\langle x \rangle$ and the variance $\langle (\Delta x)^2 \rangle$ as a function of D and t.

   (d) Show that the quantity $\theta = \int_{-\infty}^{+\infty} dx (\frac{\partial P}{\partial x})^2 \geq 0$ decreases monotonically with time and that $d\theta/dt \leq 0$. Explain the implication of this result on the character of the diffusion process.

2. **Thermodynamics of type-I superconductor**

The internal energy density per unit volume of an isotropic normal metal inside which there is a uniform magnetic flux density (or "magnetic induction") B is modeled as

$$u_{\mathrm{n}}(s, B) = u_{\mathrm{n}}^0 + \tfrac{1}{2}\gamma^{-1}s^2 + \tfrac{1}{2}\mu^{-1}B^2,$$

where the normal state parameter γ (related to specific heat) and the magnetic permeability μ are taken to be temperature-independent constants, and s is the entropy density. (This expression includes the energy density of the electromagnetic fields inside the sample.) Recall that in this system, the first law of thermodynamics has the form

$$du = Tds + HdB$$

where T is the temperature, and H is the (externally-applied) magnetic field. (We assume the volume of the system is held fixed.)

(a) Obtain the ("magnetic Gibbs") free energy density $g_{\mathrm{n}}(T, H)$ for the normal metal with externally-fixed temperature T and externally-applied magnetic field H.

At low temperatures $T < T_{\mathrm{c}}$ and small externally-applied magnetic field $H < H_{\mathrm{c}}(T)$, the normal state of the metal is unstable against (type-I) superconductivity, where the magnetic flux is completely expelled from the interior of the sample (Meissner effect), forcing B to vanish. If the superconductor is heated in the absence of an externally-applied magnetic field ($H = 0$), it becomes normal at temperature $T = T_{\mathrm{c}}$ in a continuous ("second-order") transition at which no latent heat is evolved.

The specific heat per unit volume of the superconducting state is modeled as

$$c_{\mathrm{s}} = \alpha T^3.$$

(b) What is the critical temperature T_{c}?

(c) At temperature $T = 0$, what is the applied magnetic field H_{c}^0 above which the normal state is the ground state of the metal?

(d) Obtain the free energy $g_{\mathrm{s}}(T, H) \equiv g_{\mathrm{s}}(T)$ of the superconducting state (it is independent of H).

(e) Obtain the phase boundary $H_{\mathrm{c}}(T)$ that separates the normal and superconducting regions of the $T - H$ phase diagram, and sketch this phase diagram. (Indicate any features you think are important on it, *e.g.*, first *vs.* second order transition, slope as $T \to 0$, *etc*).

3. **Sap rising in trees**

Trees can grow to enormous heights, and yet water still must be brought from the roots all the way to the leaves at the top. How do trees do this? Here we present a toy version of the problem to see if passive/diffusive transport of the water is a viable mechanism. Imagine an idealized tree growing out of soil (with water vapor concentration C_{soil}) into a well mixed atmosphere, everywhere with constant temperature $T = 300$ K and water vapor concentration $C_{air} < C_{soil}$. Our tree has pure liquid water inside, which is only in contact with the soil through the roots at height $z = 0$ and is in contact with the atmosphere through the leaves at the top of the tree at height $z = h$.

(a) Say that if the tree has height h_{eq} it is in diffusive equilibrium with the water vapor in the soil *and* the water vapor in the air. Qualitatively, what happens if the tree is shorter than h_{eq}? Taller? Why is h_{eq} the maximum height the tree could grow?

(b) To solve for the diffusive equilibrium conditions we will need to first solve for the chemical potential μ of the water vapor (which we will approximate as an ideal gas). Recall that for an ideal gas with only one particle, ignoring internal degrees of freedom, the partition function is $Z_1 = C_0 V$ where $C_0 = \left(\frac{2\pi}{\beta m}\right)^{3/2}$. What is the partition function for N indistinguishable particles Z_N? Calculate the free energy $F(N) = -k_B T \log Z$.

(c) Recalling that $dE = TdS - pdV + \mu dN$ and $F = E - TS$ show that the chemical potential $\mu = \left(\frac{\partial F}{\partial N}\right)_{T,V}$. Calculate $\mu(C)$ for an ideal gas where concentration $C = N/V$.

(d) With the expression for $\mu(C)$ for an ideal gas in hand, we can now return to our idealized tree. Now, the maximum concentration of water dissolved in the atmosphere depends heavily on temperature. Let us denote the saturating concentration of water at $T = 300$ K as C_{sat}. The soil is saturated with water, so the concentration of water vapor in the soil is $C_{soil} = C_{sat}$ while the atmosphere has constant water vapor concentration of $C_{air} = 0.9 C_{sat}$. What is the maximum height h_{eq} the tree can grow only using the difference in water vapor concentration between the soil and the air to pull the water up to its leaves? Derive an expression for h_{eq} in terms of the available variables, and give an order of magnitude estimate using the constants below.

Some useful constants:
At 300 K: $k_B T = 4.14 * 10^{-21}$J. For a water molecule: $mg = 2.60 * 10^{-25}$ N.